

The Type L hypothesis is unnecessary for topological amenability of sub-semihypergroups

Candidate full solution for arXiv:2211.01479

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Status. Candidate full solution, likely valid; human expert review is required.

1 Source question

The source is C. Bandyopadhyay, *Topological Amenability of Semihypergroups*, arXiv:2211.01479, subsequently published in *Forum Mathematicum* 36 (2024), no. 4, 943–954 [1]. The final remark on page 16 of the arXiv PDF asks whether the Type L assumption in Theorem 3.5 can be removed.

trivially that any non-commutative semihypergroup $(K, *)$ where the convolution is given as $p_x * p_y = p_y$ for each $x, y \in K$, is topological left amenable .

As mentioned at the beginning of this section, although topological left amenability of a semihypergroup immediately implies left amenability, it is still an open question whether the converse is true for general semitopological semihypergroups. Moreover, although Theorem 3.5 extends the results of [21] even for the case of locally compact (semitopological) semigroups, it is not known whether in the setting of a semihypergroup, the restriction that H be a *Type L* sub-semihypergroup, can be lifted.

Transcription of the target sentence. “Moreover, although Theorem 3.5 extends the results of [21] even for the case of locally compact (semitopological) semigroups, it is not known whether in the setting of a semihypergroup, the restriction that H be a Type L sub-semihypergroup, can be lifted.”

Here “lifted” is read in context as “removed.” Theorem 3.5 states that for a closed sub-semihypergroup H of a semitopological semihypergroup K , the following are equivalent: H has a topological left-invariant mean (TLIM), and there is a topological left H -invariant mean m on $M(K)^*$ with $m(\chi_H) = 1$. Under the additional Type L hypothesis, the source also proves equivalence with the formally weaker condition $m(\chi_H) > 0$.

2 Definitions and notation

We follow the source. For a locally compact space X , $M(X)$ is the Banach lattice of complex regular Borel measures, $P(X)$ is its set of probability measures, and convolution is written $*$. If $H \subseteq K$ is closed, μ_H denotes restriction of $\mu \in M(K)$ to H , while σ^e denotes extension by zero of $\sigma \in M(H)$ to K . The mass functional of H is

$$\chi_H(\mu) = \mu(H) \quad (\mu \in M(K)).$$

For $\rho \in M(K)$ and $F \in M(K)^*$,

$$(L_\rho F)(\mu) = F(\rho * \mu).$$

A mean m on $M(K)^*$ is *topological left H -invariant* if $m(L_\rho F) = m(F)$ for every $F \in M(K)^*$ and every $\rho \in P(K)$ with $\text{supp } \rho \subseteq H$.

Define the positive restriction map

$$e_H : M(H)^* \longrightarrow M(K)^*, \quad e_H(\phi)(\mu) = \phi(\mu_H).$$

3 Proof Intuition

The source proof estimates how restriction fails to commute with left convolution and invokes Type L to rewrite an inverse-set expression. There is a more direct description of the error. If the left factor is supported in H , then the part of the right-hand measure already in H stays in H , because $H * H \subseteq H$. Thus the entire restriction defect is precisely the mass that starts in H^c and is transported into H .

This incoming mass is measured by the positive functional $L_{\nu^e} \chi_H - \chi_H$. The restriction defect is dominated in lattice modulus by a scalar multiple of that functional. But an H -invariant mean annihilates it. Consequently the normalized restriction of any invariant mean assigning positive mass to H is automatically a TLIM on H .

4 Result

Theorem 1 (Theorem 3.5 without Type L). *Let K be a semitopological semihypergroup and let $H \subseteq K$ be a closed sub-semihypergroup. The following are equivalent:*

- (1) *H admits a topological left-invariant mean;*
- (2) *there is a topological left H -invariant mean $m \in \mathcal{M}(K)$ such that $m(\chi_H) = 1$;*
- (3) *there is a topological left H -invariant mean $m \in \mathcal{M}(K)$ such that $m(\chi_H) > 0$.*

In particular, the Type L hypothesis in the final part of the source's Theorem 3.5 is unnecessary.

Proof. The implications (1) $\Rightarrow$ (2) $\Rightarrow$ (3) are already contained in the source and do not use Type L. We prove (3) $\Rightarrow$ (1).

Let m be as in (3), set $\alpha = m(\chi_H) > 0$, and define

$$m_0 = \alpha^{-1} m \circ e_H \in M(H)^{**}.$$

The map e_H is positive. If $\mathbf{1}_H \in M(H)^*$ is the total-mass functional, then $e_H(\mathbf{1}_H) = \chi_H$, and hence $m_0(\mathbf{1}_H) = 1$. Thus m_0 is a mean on $M(H)^*$.

Fix $\nu \in P(H)$ and write $\bar{\nu} = \nu^e \in P(K)$. Introduce the positive operator

$$Q_\nu : M(K) \longrightarrow M(H), \quad Q_\nu(\mu) = (\bar{\nu} * (\mu_{H^c})^e)_H.$$

For every $\phi \in M(H)^*$, bilinearity of convolution and $H * H \subseteq H$ give the exact identity

$$e_H(L_\nu \phi) - L_{\bar{\nu}} e_H(\phi) = -\phi \circ Q_\nu. \tag{1}$$

Indeed, for $\mu \in M(K)$ we decompose $\mu = (\mu_H)^e + (\mu_{H^c})^e$. The measure $\bar{\nu} * (\mu_H)^e$ is supported in H , and its restriction to H is $\nu * \mu_H$. Subtracting this term from $(\bar{\nu} * \mu)_H$ leaves exactly $Q_\nu(\mu)$.

Now set

$$g_\nu = L_{\bar{\nu}} \chi_H - \chi_H \in M(K)^*.$$

For every $\mu \in M^+(K)$, the part $(\mu_H)^e$ stays in H after convolution by the probability measure $\bar{\nu}$, with unchanged total mass. Therefore

$$g_\nu(\mu) = (\bar{\nu} * \mu)(H) - \mu(H) \quad (2)$$

$$= (\bar{\nu} * (\mu_{H^c})^e)(H) = \|Q_\nu(\mu)\| \geq 0. \quad (3)$$

Thus g_ν is positive. Since Q_ν is positive, the usual variation inequality gives $|Q_\nu(\mu)| \leq Q_\nu(|\mu|)$. Hence, in the dual Banach lattice $M(K)^*$,

$$|\phi \circ Q_\nu| \leq \|\phi\| g_\nu. \quad (4)$$

Because $\text{supp } \bar{\nu} \subseteq H$, topological left H -invariance of m implies

$$m(g_\nu) = m(L_{\bar{\nu}}\chi_H) - m(\chi_H) = 0.$$

Positivity of m and (4) now yield

$$|m(\phi \circ Q_\nu)| \leq m(|\phi \circ Q_\nu|) \leq \|\phi\| m(g_\nu) = 0.$$

Applying m to (1) and using H -invariance once more, we get

$$\begin{aligned} m_0(L_\nu\phi) - m_0(\phi) &= \alpha^{-1}m(e_H(L_\nu\phi) - e_H(\phi)) \\ &= \alpha^{-1}m(e_H(L_\nu\phi) - L_{\bar{\nu}}e_H(\phi)) \\ &\quad + \alpha^{-1}m(L_{\bar{\nu}}e_H(\phi) - e_H(\phi)) = 0. \end{aligned}$$

This holds for every $\nu \in P(H)$ and $\phi \in M(H)^*$, so m_0 is a TLIM on H . $\square$

5 Verification, scope, and novelty

The proof uses only positivity, bilinearity, $H * H \subseteq H$, and topological H -invariance. It does not use the Type L identity, an involution, an identity element, commutativity, or any hypergroup axiom. The crucial point is the exact defect formula (1); the Banach-lattice estimate then turns the defect into a positive mean-zero quantity.

This result answers only the Type L question. It does not address the other question in the same source remark, namely whether ordinary left amenability implies topological left amenability for general semitopological semihypergroups.

A bounded novelty search covered the run's registry and packet indexes, exact phrases from the source question, the local arXiv corpus, the source's 2024 publication record, the author's later arXiv:2404.18261 and 2024 amenability survey metadata, and web searches combining “Type L”, “sub-semihypergroup”, and “topological left H -invariant mean”. No later answer was found. This does not establish originality.

Human-review recommendation. Review at high priority. The main checks are the dual Banach-lattice modulus step in (4) and agreement of the source's complex-measure conventions with the restriction operator Q_ν . A specialist citation search should also check whether the same incoming-mass argument appears under different terminology.

References

- [1] C. Bandyopadhyay, *Topological Amenability of Semihypergroups*, arXiv:2211.01479; Forum Math. **36** (2024), no. 4, 943–954.